

Reading-H T6 entry package: the candidate conditional theorem, its complete hypothesis set, and the assembly chain

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

Operator verdict #14 flipped gate STEP-5B to CLOSED-CONDITIONAL (margins $\times 59.4$ / $\times 2.6$ floors after AddE), and STEP-5B was the registered “gateway for any whole-Reading-H T6 discussion” (claims/GATES.md, ROADMAP Stage 2). This note opens that discussion by assembling the candidate conditional theorem, enumerating its COMPLETE hypothesis set with tracking locations, and recording the promotion proposal. Per governance (no self-promotion), the B1 tier field is NOT changed by this note: the proposal awaits operator sign-off.

2. The candidate T6 theorem

Candidate Theorem (Reading-H selection, conditional). At the corrected canonical operating point $r_{\text{braz}} = \mu^2 = 0.005$, for every pattern $\mathcal{P}$ in the H-ADM-COH-amended admissible competitor class, at the certified anchor intensities $I \in \{4 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$,

$$\Delta F[\mathcal{P}] = F_{\text{TECT}}[\mathcal{P}] - F_{\text{TECT}}[\mathcal{R}_H] > 0,$$

CONDITIONAL on the hypothesis set $\{\text{H-LAYER}^b, \text{H-A0}, \text{H-ADM-COH}, \text{SC-SCOPE}\}$ enumerated in section 3, where H-LAYER^b denotes the layer hypothesis with its beyond-layer residual DISCHARGED on the amended class by the STEP-5B closure.

Assembly chain (every link on disk and operator-reviewed):

1. **Layer-level positivity (B2, T6 on {H-LAYER, H-A0}).** Proposition A (Math437 v1.2, certified by Math442 dual audit + operator sign-off): for every admissible pattern and every PD trial, $F_{\text{var}}(t, m) > F_{\text{var}}(0, m^*) = F_{\text{var}}[\mathcal{R}_H]$, with the four-interval quadrature-free floor chain giving the worst-case band margin $+0.00432$ (intervals: $+0.00432$ / $+0.00417$ / $+0.00467$; the binding one is the band).
2. **Beyond-layer domination (B5 chain, STEP-5B CLOSED-CONDITIONAL, verdict #14 + AddE polish).** For every pattern of the H-ADM-COH-amended class, at the three anchor intensities,

$$|\delta F_{\text{off}}(\mathcal{P})| \leq \frac{K(n_{\text{pack}})(\lambda' I)^2 J_{\text{eff}}}{4(1 - a_0)} \leq \frac{m_{\text{band}}}{2.6} \quad (m_{\text{band}} = 0.00432; \text{ floor at } I = 2 \times 10^{-3}),$$

with the production anchor at $m_{\text{band}}/59.4$ (AddE hardened floors; band tops $\times 290.9$ / $\times 12.7$).

3. **Composition.** For every amended-class $\mathcal{P}$ at the certified intensities,

$$\Delta F[\mathcal{P}] \geq m_{\text{band}} - \frac{m_{\text{band}}}{2.6} = m_{\text{band}} \left(1 - \frac{1}{2.6}\right) = 0.00432 \times 0.6154 \simeq +2.66 \times 10^{-3} > 0,$$

i.e. the composed worst-case margin is $\sim 62\%$ of the layer floor at the thinnest certified endpoint, and $\geq 0.00432 (1 - 1/59.4) \simeq +4.25 \times 10^{-3}$ at the production anchor.

The composition is valid because the two links live at the SAME bookkeeping order (matched second-cumulant; SC-SCOPE below), the same dressed kernel and anchor calibration (A1-KERNEL-CONV convention; $\hat{r}$, $\lambda' = 8.0551$, J machinery shared), and the STEP-5B budget $m_{\text{STEP-5B}}$ was DEFINED against the Prop-A band margin $+0.00432$ throughout the B5 chain (artefact field `anchor.margin` = 0.00432).

3. The complete hypothesis set (registry)

Every hypothesis the candidate theorem rests on, with its tracking location and discharge path. Nothing is hidden in the assembly.

Hypothesis	Content (binding form)	Tracking / discharge
H-LAYER ^b	The comparison layer is the isotropic Gaussian–Hartree variational layer; its beyond-diagonal residual (formerly “unexecuted for non-enumerated patterns”) is DISCHARGED on the amended class by the STEP-5B closure. The superscript ^b marks exactly this discharged form: what remains of H-LAYER is the layer framing itself (Gaussian–Hartree variational comparison as the free-energy bookkeeping), not the beyond-layer exhaustiveness.	claims/GATES.md (H-LAYER row); residual was STEP-5B (now CLOSED-CONDITIONAL, verdict #14); the layer-framing remainder is inventoried in the companion residual note (next unit).
H-A0	$A = 0$ uniqueness and zero-at-gap structure certified numerically on a consistent quadrature (3.1×10^{-5} internal convergence; 5.5×10^{-3} scheme-gap offset recorded as a measure-convention systematic). Penalty floors P_B are quadrature-free and do NOT rest on it.	claims/GATES.md (H-A0 row); discharge = quadrature-scheme unification or an analytic $A = 0$ uniqueness proof (audit note planned as a separate unit).
H-ADM-COH	Amended admissible-competitor definition (verdict #14): finite reading sets with pairwise separations $\geq \theta_{\min}(\hat{r}) = \sqrt{\hat{r}}/(2q_0^2\sqrt{C})$; sub-resolution mass belongs to the Gaussian sea. Energy-faithful by the AddC indistinguishability lemma ($c_{\text{ind}} = 30.1$; ratios $\times 898/ \times 139/ \times 33$) and depth-free by the AddE cross pin ($c_{\text{total}} \leq 20$).	claims/GATES.md STEP-5B row (verdict #14 verbatim); B5 notes AddB/AddC/AddD/AddE; unrestricted-class discharge = DR-2 (research-grade, off critical path).
SC-SCOPE	Matched second-cumulant bookkeeping: every B5 bound and the Prop-A comparison live at second-cumulant order with sextic corrections folded multiplicatively (Lemma H, $\varepsilon_4 \leq 0.16$ on the closed region); third-cumulant vertex terms are NOT bounded class-wide.	B5 main chain (G2 closed at this level, Lemmas H/I/J); the third-cumulant lift assessment is registered as a follow-up unit.

Additionally the theorem inherits the standing convention dependency A1-KERNEL-CONV (production-kernel convention, T5) and the anchor calibration provenance (Math434 chain via archive) — these are DEPENDENCIES, not hypotheses: they fix the operating point at which the statement is made, and the statement is scoped to that point.

4. What the candidate T6 statement does NOT include

For the no-overclaim record, the following are explicitly OUTSIDE the candidate theorem and remain tracked by their own gates:

- **Estimator-grade ΔF figures** (B1’s enumerated-race values): the class-wide positivity above does not certify the NUMERICAL ΔF values of the enumerated races beyond estimator grade; ESTIMATOR-UPGRADE (GAP-2 instance) stays OPEN. Note the logical direction: the enumerated races are now corroborating evidence for a statement whose load-bearing chain is Prop-A + STEP-5B, both analytic-with-certified-constants.
- **Off-anchor intensities**: the margin chain is certified at the three production anchors (I from 4×10^{-4} down to 2×10^{-3}); intermediate intensities inherit plausibility from the monotone structure of the chain (K budgets shrink, $c_{\text{ind}} I^2$ grows quadratically) but are not separately certified. An interval-arithmetic sweep is a registered polish item.
- **μ^2 -neighbourhood**: ROBUSTNESS-MU2 stays OPEN; the statement is pinned to $\mu^2 = 0.005$.
- **Higher shells / anisotropic harmonics**: G3PB-III stays OPEN (single-shell amended class plus the two-shell floor Lemma J at $\times 1.70$; the general multi-shell statement is not part of this theorem).
- **Third-cumulant order**: SC-SCOPE fences the bookkeeping; the lift assessment is a follow-up unit.

5. Numerical verification

No NEW machine numbers are introduced by this note. Every constant used is already certified in the existing artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#) (script [codes/vacuum/beyond_layer_gershgorin_bound.py](#) v1.14.0, 192/192 self-test asserts PASS) and in the migrated Prop-A record ([archive/legacy/notes/Math437/...v1.2](#), floors $P_B(M_+) = 0.0052460$, $P_B(M_c) = 0.0092199$, band sum $+0.00432$; Math442 certification). The single new arithmetic step is the composition

$$0.00432(1 - 1/2.6) = 0.00432 \times 0.61538 = 2.658 \times 10^{-3}, \quad 0.00432(1 - 1/59.4) = 4.247 \times 10^{-3},$$

which is exact closed-form arithmetic on certified constants.

Quantitative sanity check (mandatory): dimensional — both sides of the composition are free-energy densities in the same canonical units as the $+0.00432$ band margin (pure subtraction, no unit mixing); magnitude — the composed endpoint margin 2.66×10^{-3} is 62% of the layer floor, consistent with $1/2.6 = 0.385$ being subtracted; limit case — as the STEP-5B margin ratio $\rightarrow \infty$ the composed bound recovers the full Prop-A floor $+0.00432$, and at ratio $= 1$ it degenerates to 0 (the composition fails exactly when the gate margin is exhausted, as it must); sign-direction — δF_{off} enters with absolute value, so the bound holds whether the beyond-layer correction helps or hurts the competitor.

6. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) “The composition double-counts or mis-aligns the two margins: Prop-A’s $+0.00432$ is a layer-level floor at the variational comparison, while the STEP-5B budget was tuned against the same

number — is the subtraction legitimate?” DISMISSED with the chain’s own construction: the B5 budget $m_{\text{STEP-5B}}$ was DEFINED as the Prop-A band margin from the first note of the chain (artefact `anchor.margin = 0.00432`), precisely so that “ $|\delta F_{\text{off}}| < m_{\text{STEP-5B}}$ ” means “the beyond-layer correction cannot exhaust the layer floor.” The subtraction $m(1 - 1/\rho)$ with ρ the certified margin ratio is the intended composition, at matched bookkeeping order, same kernel, same anchors. No constant is used twice in the same inequality.

- (β) “*H-LAYER is listed as discharged ($\flat$), but the amendment H-ADM-COH redefined the class — so the discharge holds only on the amended class, and quietly promoting that to ‘H-LAYER discharged’ is an overclaim.*” VALID-with-mitigation: agreed, and the note’s notation is built around exactly this point — H-LAYER ^{$\flat$} is defined as “discharged ON THE AMENDED CLASS,” the amendment itself is a separate listed hypothesis (H-ADM-COH), and the candidate theorem quantifies over the amended class only. The unrestricted-class statement remains OPEN (DR-2, research-grade) and is excluded from the theorem by construction. The residual inventory between “amended-class conditional” and “unconditional” is the next unit’s content.
- (γ) “*The thin endpoint: at $I = 2 \times 10^{-3}$ the composed margin rests on the $\times 2.6$ FLOOR, which itself rests on the criterion-band conservative end and the $n_{\text{pack}} = 16/\theta_{\text{min}}^2$ packing constant. A T6 entry on a $\times 2.6$ link is fragile.*” VALID-with-mitigation: quoted honestly — the endpoint is the weakest certified link and is flagged as such here and in AddE; the band top at that intensity is $\times 12.7$ and the production anchor floor is $\times 59.4$. The T6-conditional tier does not require margin comfort, only a valid conditional proof with explicit hypotheses; the thinness is a robustness concern tracked by the registered interval-sweep polish, not a gap in the conditional chain.
- (δ) “*B1’s evidence grade is ESTIMATOR + EXECUTED; promoting B1 to T6 on the strength of B2 + B5 changes what the claim IS (from enumerated races to class-wide selection) — a tier change disguised as a statement change.*” VALID-with-mitigation: correct that the T6 object is a STRONGER statement than the original enumerated-race claim; this is why the note proposes the promotion as an explicit operator decision with the new statement text supplied (section 7), rather than editing the tier in place. If the operator prefers, the alternative bookkeeping is a NEW claim card (e.g. B6-RH-CLASSWIDE) holding the conditional theorem at T6 while B1 stays T5 for the enumerated races; both options are workable and the choice is the operator’s. This note records the proposal in `status.json` scope/notes ONLY.

7. Proposal record (operator decision required)

- **Proposed action:** B1-RH-ENUM tier T5 $\rightarrow$ T6-CONDITIONAL, with the statement upgraded to the candidate theorem of section 2 and the hypothesis field set to {H-LAYER ^{$\flat$} , H-A0, H-ADM-COH, SC-SCOPE} (the latter two to be added as named-hypothesis rows in `claims/GATES.md` upon sign-off).
- **Alternative:** a new claim card B6-RH-CLASSWIDE at T6-CONDITIONAL carrying the assembled theorem; B1 stays T5 for the enumerated-race content. (Operator’s choice; the assembly chain is identical either way.)
- **Not proposed:** any change to B2 (stays T6 on {H-LAYER, H-A0}), B5 (stays T5-CANDIDATE per verdict #14), or any gate row.

8. Result footer

Result ID: B1-RH-ENUM / reading-h-t6-entry (mainline)
Precise statement: CANDIDATE T6 THEOREM (assembled, proposal):
Delta F[P] > 0 for every pattern P of the
H-ADM-COH-amended admissible class at
 $\mu^2 = 0.005$, I in {4e-4, 1e-3, 2e-3},
CONDITIONAL on {H-LAYER-flat, H-A0,
H-ADM-COH, SC-SCOPE}. Composed worst-case
margin = $0.00432 \cdot (1 - 1/2.6) \sim +2.66e-3$
(endpoint floor); +4.25e-3 at the anchor.
Scope: Amended class only; three certified anchor
intensities; operating point pinned;
second-cumulant bookkeeping. Estimator-grade
enumerated Delta-F figures NOT upgraded
(ESTIMATOR-UPGRADE stays OPEN).
Dependencies: B2-PROPA-HLAYER (T6: Prop A), B5 chain
(STEP-5B CLOSED-CONDITIONAL, verdict #14 +
AddE), A1-KERNEL-CONV, anchor calibration.
Evidence grade: ANALYTIC (assembly of certified links) --
no new machine numbers; one exact arithmetic
composition on certified constants.
Reproduction command: python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output: claims 192/192 PASS (existing artefact cited;
no new script introduced by this note).
Falsification gate: Any amended-class pattern with Delta F ≤ 0 at
a certified intensity; failure of any cited
link (Prop-A floor chain, STEP-5B margin
chain) under audit; discovery that the B5
budget and the Prop-A floor are NOT the same
 $m_{\text{band}} = 0.00432$ (would break the
composition).
Tier before / after: B1 stays T5 (tier field UNCHANGED by this
note). PROPOSED: T5 -> T6-CONDITIONAL
(or new card B6-RH-CLASSWIDE at T6-COND.) --
OPERATOR SIGN-OFF REQUIRED.
No-overclaim statement: This is a PROPOSAL note. No tier, gate, or
hypothesis row is changed by it. The theorem
is conditional and scope-fenced; the
unrestricted class, off-anchor intensities,
 μ^2 neighbourhood, higher shells, and
third-cumulant order are all explicitly
excluded. PDF build + linter execution are
PENDING OPERATOR-SIDE (sandbox shell
unavailable to this agent session).
Next required action: OPERATOR DECISION on the promotion form
(in-place T6-CONDITIONAL vs new B6 card);
then: H-A0 removal-pathway audit, H-LAYER

residual inventory, SC-SCOPE third-cumulant
lift assessment (registered as the next
units).